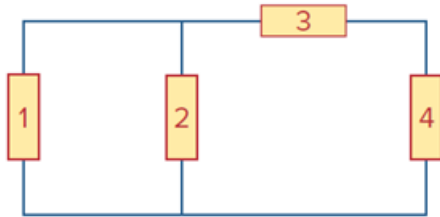


Chapter 1, Problem 17P

Problem

Figure 1.28 shows a circuit with four elements, $p_1 = 60 \text{ W}$ absorbed, $p_3 = -145 \text{ W}$ absorbed, and $p_4 = 75 \text{ W}$ absorbed. How many watts does element 2 absorb?

Figure 1.28:



Step-by-step solution

Step 1 of 2

Consider the textbook Figure 1.28. Element p_1 absorbs 60 W , element p_3 absorbs -145 W and element p_4 absorbs 75 W . The algebraic sum of power in any electric circuit at any instant time is zero.

Step 2 of 2

Calculate the power absorbed by element p_2 .

$$p_1 + p_2 + p_3 + p_4 = 0$$

$$60 + p_2 - 145 + 75 = 0$$

$$p_2 = 10 \text{ W}$$

Therefore, the power absorbed by element p_2 is 10 W.